Hirmay Sandesara

Computational science, tensor networks, and quantum simulation

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Research Interests

Tensor-network algorithms and efficient numerical methods for quantum many-body systems, with a focus on representations that avoid explicit exponential storage. Particularly interested in identifying regimes and algorithmic approaches for which tensor-network ranks remain manageable, and in developing scalable methods for many-body simulation.

Education

Technical University of Munich — Munich, Germany
M.Sc. Computational Science and Engineering, quantum computing focus — Oct 2024 – present

- Current grade: **1.5** (German scale; approximately 3.65/4.0).
- **TUM merit-based tuition waiver**; **Deutschlandstipendium**; CSE student representative.

Ahmedabad University — Ahmedabad, India
B.Tech. Computer Science and Engineering — Jul 2019 – May 2023

- Grade: 1.4 (German equivalent; 3.73/4.0); ranked **2nd of 140 students**.
- Dean's Recognition List; graduated with a certificate of highest excellence.

Selected Publications and Research Output

H. Sandesara, B. Decker, and C. Mendl. "MPS-Compressed Majorana Propagation with Virtual $\mathbb{Z}_2$ Charges." Manuscript in preparation.

H. Sandesara, A. Shukla, and P. Vedula. "A quantum approach for optimal control." *Quantum Information Processing* **24**, 95 (2025). doi:10.1007/s11128-025-04710-z.

H. Sandesara et al. "Space Sustainability Using Quantum Computing." White paper, OSF Preprints (2022). doi:10.31219/osf.io/wndfb.

Research Experience

Technical University of Munich — Munich, Germany
Master's thesis in tensor-network algorithms; supervisor: Prof. Christian B. Mendl — Current

- Developing an operator-space matrix product state (MPS) approach for propagating Majorana operators in fermionic systems with long-range interactions.
- Introducing virtual $\mathbb{Z}_2$ charge labels on the MPS bonds to record accumulated fermionic parity, with the local rule $q_R = q_L \oplus p(s)$.
- Handling signs from long Jordan–Wigner and Majorana strings through endpoint and virtual-state information while preserving charge-aware tensor contractions and SVD compression.

Ahmedabad University — Ahmedabad, India
Research Assistant — Sep 2023 – May 2024

- Developed a multidimensional formulation for nonlinear optimal-control problems using constrained Hamiltonian dynamics and canonical quantization.
- Designed and implemented a variational quantum method for the resulting broken-PT-symmetric, non-Hermitian systems using Python and Mathematica.
- Evaluated control problems with PennyLane, Qibo, SciPy, symbolic computation, parallel integration, and PDE solvers; work published in *Quantum Information Processing*.

Quantum Open Source Foundation — Remote
Mentee; mentor: Ryan Hill, qBraid — Apr 2024 – Aug 2024

- Selected as one of approximately 40 participants from more than 200 applicants.
- Worked on the design of a Python framework for configurable quantum reservoir-computing models and integration with QuEra computing resources.

Santa Fe Institute Remote
Student Researcher Jan 2024 – Apr 2024

- Collaborated on "Darkening Skies," a project studying how the COVID-19 period affected ecosystems, conservation funding, and conservation efforts.

Artificial Brain Remote
Quantum Computing Intern Jul 2022 – Dec 2022

- Co-authored a white paper formulating space-debris removal as knapsack and routing optimization problems and surveying QAOA and quantum-annealing approaches.
- Contributed to quantum scheduling for Earth-observation satellites and to a D-Wave portfolio-optimization prototype targeting the Sharpe ratio.

Selected Technical Projects

Entanglement distillation on a constrained quantum network 2026
MIT iQuHACK, IonQ Challenge – 2nd prize

- Led a four-person team that designed a heuristic parity-check protocol for entanglement distillation under a minimal single-Bell-pair budget.
- Achieved a challenge score of 89 and claimed network edges at fidelity at least 0.9; see the code and technical report.

NISQ methods for open-system simulation 2024
QHack Open Hackathon – 3rd prize, Seeing the Future; AWS Top 3

- Led development of shallow parameterized circuits for state preparation and time evolution in open quantum systems.
- Tested the state-preparation routine on synthetic inputs and Alzheimer's patient data using parallel GPU optimization with NVIDIA cuQuantum; also explored logical cat-qubit simulations. Code and report.

Hamiltonian simulation of coupled classical oscillators 2024
Classiq QRISE Challenge – winning submission

- Derived the Hamiltonian and implemented state preparation and simulation workflows for a coupled-oscillator model on Classiq's platform.
- Compared construction choices with the practical aim of maintaining fidelity while reducing circuit depth. Project repository.

Zero-noise extrapolation 2024
Quantum Open Source Foundation screening task

- Implemented depolarizing-noise models, unitary folding, and several extrapolation rules for Qiskit circuits; benchmarked estimates against the zero-noise limit. Implementation.

Honors and Awards

2nd Prize, MIT iQuHACK IonQ Challenge 2026
3rd Prize, QHack Seeing the Future Challenge; AWS Top 3 2024
Winner, QRISE Classiq Challenge 2024
TUM merit-based tuition waiver and Deutschlandstipendium 2024
Rank 2/140, Ahmedabad University; Dean's Recognition List 2023

Open Source, Mentoring, and Service

- Contributed an implementation of uniform-superposition state preparation to Qiskit's circuit library; co-authored material on Grover's algorithm for the Qiskit Textbook; and helped fix a Qiskit Terra bug.
- Translated more than 5,000 words of Qiskit educational material into Hindi.
- Mentored participants during the IBM Quantum Challenges in 2022 and 2024 by debugging tasks and explaining relevant concepts.
- Organized quantum-computing workshops and hackathons; co-founded Qtangled, whose first introductory workshop reached 250 participants.

Technical Skills

Numerical methods	Matrix product states/operators, tensor contractions, SVD and truncation, numerical linear algebra, optimization, PDE solvers, quantum simulation
Programming	Python, Julia, C, C++, MATLAB, Mathematica, SQL
Libraries and tools	NumPy, SciPy, SymPy, Qiskit, PennyLane, Qibo, QuTiP, cuQuantum, Matplotlib, PyTest, Git, LaTeX